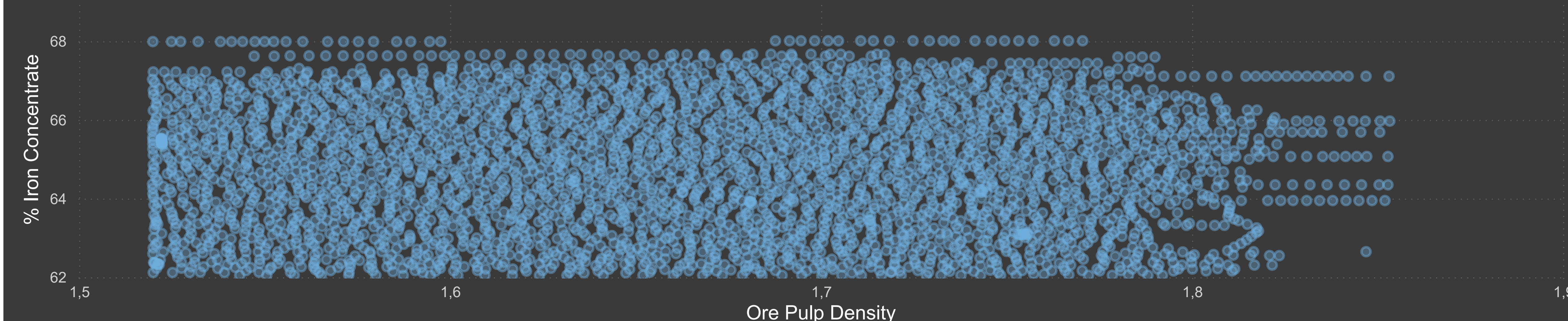
Mining Flotation Process Analytics Dashboard

Machine Learning & Process Optimization Analysis

Key Performance Indicators

Avg Iron Concentrate	Avg Silica Concentrate	R^2 Score	MAE
65,05	2,33	0,96	0,09

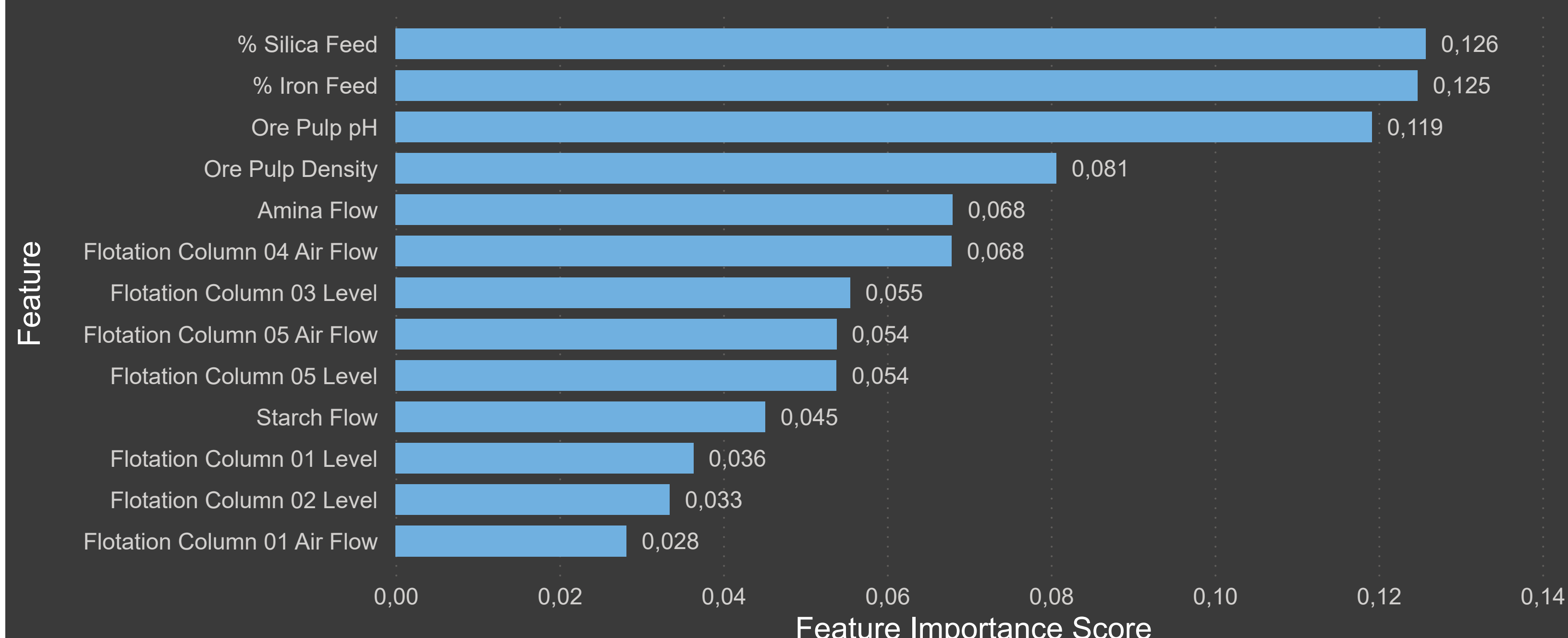
Ore Pulp Density vs Iron Concentrate



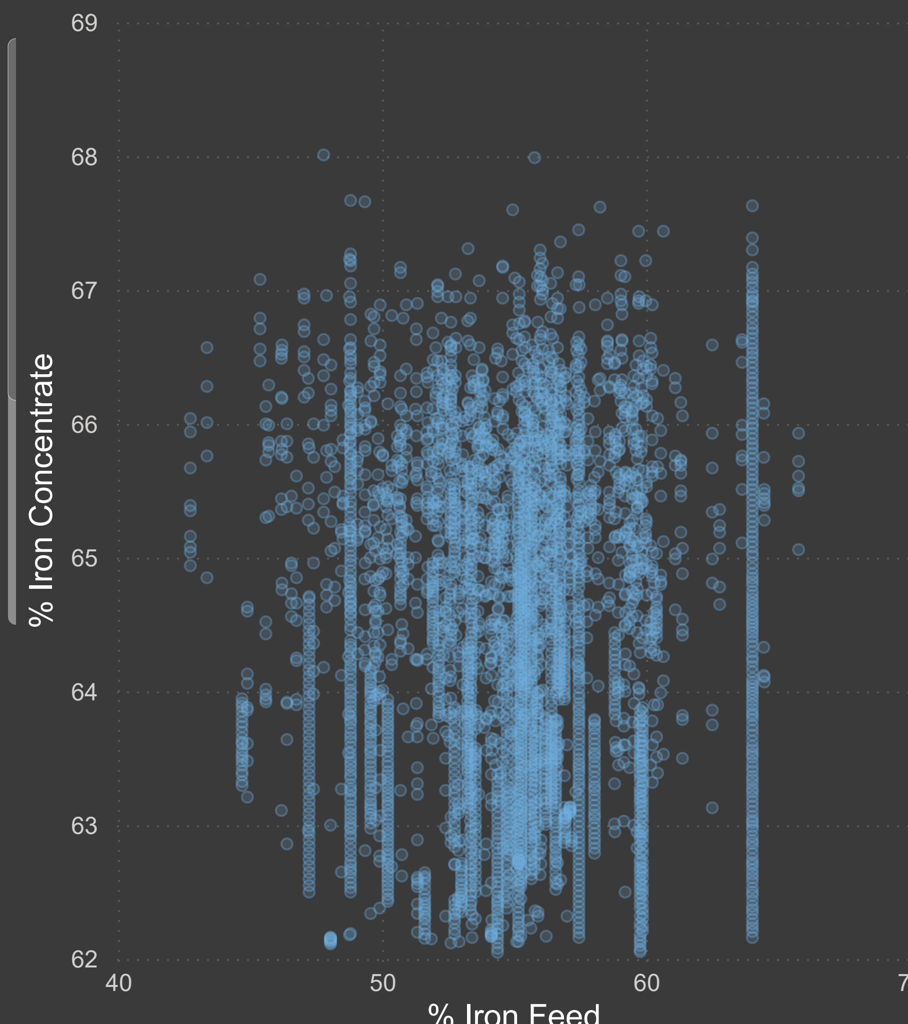
Process Analysis Dashboard

Feature importance analysis indicates that iron feed percentage, ore pulp density and flotation variables are the main drivers affecting iron concentrate production. Operational relationships observed in the process variables suggest opportunities for optimization and process stabilization through data-driven monitoring.

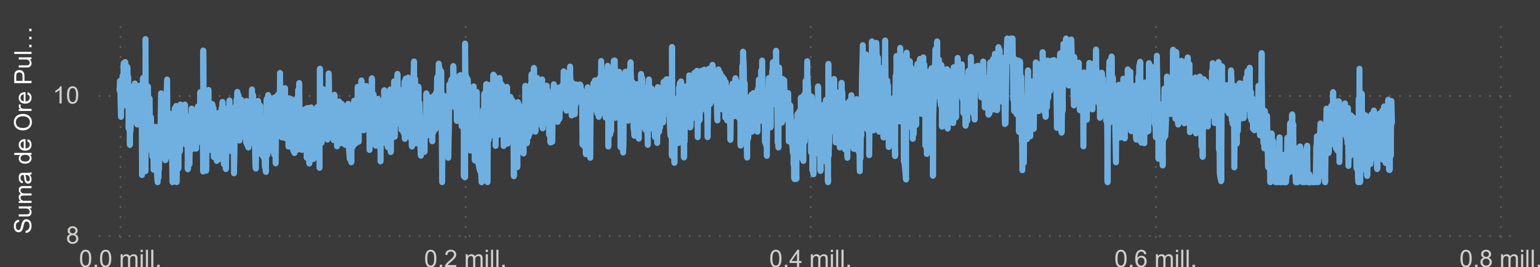
Feature Importance - Random Forest Model



% Iron Feed y % Iron Concentrate



Ore Pulp pH Trend



Machine Learning Performance Dashboard

The Random Forest model achieved strong predictive performance, with an R^2 score of 0.96 and low prediction error (MAE = 0.09). Results indicate that operational variables such as iron feed percentage, ore pulp density and pH are highly influential in predicting iron concentrate levels. The analysis demonstrates the potential of machine learning and data analytics to support operational optimization in mining flotation processes.

R^2 Score	MAE
0,96	0,09

Actual vs Predicted Iron Concentrate